

-- Create the database

```
MariaDB [(none)]> create database fact_library;  
Query OK, 1 row affected (0.007 sec)
```

```
MariaDB [(none)]> use fact_library;  
Database changed  
MariaDB [fact_library]>
```

-- Create tables

```
MariaDB [fact_library]> CREATE TABLE PATRON ( Pat_ID INT PRIMARY KEY, Pat_FName VARCHAR(100) NOT NULL,  
Pat_LName VARCHAR(100) NOT NULL, Pat_Type VARCHAR(50) NOT NULL);  
Query OK, 0 rows affected (0.065 sec)
```

```
MariaDB [fact_library]> CREATE TABLE AUTHOR ( Au_ID INT PRIMARY KEY, Au_FName VARCHAR(100) NOT NULL,  
Au_LName VARCHAR(100) NOT NULL, Au_BirthYear INT NULL);  
Query OK, 0 rows affected (0.036 sec)
```

```
MariaDB [fact_library]> CREATE TABLE BOOK ( Book_Num INT PRIMARY KEY, Book_Title VARCHAR(255) NOT NULL,  
Book_Year INT NOT NULL, Book_Cost DECIMAL(10, 2) NOT NULL, Book_Subject VARCHAR(100) NOT NULL, Pat_ID INT  
NULL, FOREIGN KEY (Pat_ID) REFERENCES PATRON(Pat_ID) ON DELETE SET NULL );  
Query OK, 0 rows affected (0.045 sec)
```

```
MariaDB [fact_library]> CREATE TABLE WRITES ( Book_Num INT, Au_ID INT, PRIMARY KEY (Book_Num, Au_ID),  
FOREIGN KEY (Book_Num) REFERENCES BOOK(Book_Num) ON DELETE CASCADE, FOREIGN KEY (Au_ID) REFERENCES  
AUTHOR(Au_ID) ON DELETE CASCADE );  
Query OK, 0 rows affected (0.041 sec)
```

```
MariaDB [fact_library]> CREATE TABLE CHECKOUT ( Check_Num INT PRIMARY KEY, Book_Num INT NOT NULL, Pat_ID  
INT NOT NULL, Check_Out_Date DATE NOT NULL, Check_Due_Date DATE NOT NULL, Check_In_Date DATE NULL,  
FOREIGN KEY (Book_Num) REFERENCES BOOK(Book_Num) ON DELETE CASCADE, FOREIGN KEY (Pat_ID) REFERENCES  
PATRON(Pat_ID) ON DELETE CASCADE );  
Query OK, 0 rows affected (0.043 sec)
```

-- Populate the database

```
MariaDB [fact_library]> INSERT INTO AUTHOR VALUES (185, 'Benson', 'Reeves', 1990), (218, 'Rachel',  
'Beatney', 1983), (229, 'Carmine', 'Salvadore', NULL), (251, 'Hugo', 'Bruer', 1972), (262, 'Xia',  
'Chiang', NULL), (273, 'Reba', 'Durante', 1969), (284, 'Trina', 'Tankersly', 1961), (383, 'Neal',  
'Walsh', 1980), (394, 'Robert', 'Lake', 1982), (438, 'Perry', 'Pearson', 1986), (460, 'Connie',  
'Paulsen', 1983), (559, 'Rachel', 'McGill', NULL), (581, 'Manish', 'Aggerwal', 1984), (592, 'Lawrence',  
'Sheel', 1976), (603, 'Julia', 'Palca', 1988);  
Query OK, 15 rows affected (0.038 sec)  
Records: 15 Duplicates: 0 Warnings: 0
```

```
MariaDB [fact_library]> INSERT INTO PATRON VALUES (1160, 'robert', 'carter', 'Faculty'), (1161, 'Kelsey',  
'Koch', 'Faculty'), (1165, 'Cedric', 'Baldwin', 'Faculty'), (1166, 'Vera', 'Alvarado', 'Student'), (1167,  
'Alan', 'Martin', 'FACULTY'), (1170, 'Cory', 'Barry', 'faculty'), (1171, 'Peggy', 'Marsh', 'STUDENT'),  
(1172, 'Tony', 'Miles', 'STUDENT'), (1174, 'Betsy', 'Malone', 'STUDENT'), (1180, 'Nadine', 'Blair',  
'STUDENT'), (1181, 'Allen', 'Horne', 'Student'), (1182, 'Jamal', 'Melendez', 'STUDENT'), (1183, 'Helena',  
'Hughes', 'Faculty'), (1184, 'Jimmie', 'Love', 'Student'), (1185, 'Sandra', 'Yang', 'student'), (1200,  
'Lorenzo', 'Torres', 'Student'), (1201, 'Shelby', 'Noble', 'Student'), (1202, 'Holly', 'Anthony',  
'Student'), (1203, 'Tyler', 'Pope', 'STUDENT'), (1204, 'Thomas', 'Duran', 'Student'), (1205, 'Claire',  
'Gomez', 'student'), (1207, 'Iva', 'Ramos', 'Student'), (1208, 'Ollie', 'Cantrell', 'Student'), (1209,  
'Rena', 'Mathis', 'Student'), (1210, 'Keith', 'Cooley', 'STudent'), (1211, 'Jerald', 'Gaines',  
'Student'), (1212, 'Iva', 'McClain', 'Student'), (1213, 'Desiree', 'Rivas', 'Student'), (1214, 'Marina',  
'King', 'Student'), (1215, 'Maureen', 'Downs', 'Student'), (1218, 'Angel', 'Terrell', 'Student'), (1219,  
'Desiree', 'Harrington', 'Student'), (1220, 'Carlton', 'Morton', 'Student'), (1221, 'Gloria', 'Pitts',  
'Student'), (1222, 'Zach', 'Kelly', 'Student'), (1223, 'Jose', 'Hays', 'Student'), (1224, 'Jewel',  
'England', 'Student'), (1225, 'Wilfred', 'Fuller', 'Student'), (1226, 'Jeff', 'Owens', 'Student'), (1227,  
'Alicia', 'Dickson', 'Student'), (1228, 'Homer', 'Goodman', 'Student'), (1229, 'Gerald', 'Burke',  
'Student'), (1237, 'Brandi', 'Larson', 'Student'), (1238, 'Erika', 'Bowen', 'Student'), (1239, 'Elton',  
'Irwin', 'Student'), (1240, 'Jan', 'Joyce', 'Student'), (1241, 'Irene', 'West', 'Student'), (1242,  
'Mario', 'King', 'Student'), (1243, 'Roberto', 'Kennedy', 'Student'), (1244, 'Leon', 'Richmond',  
'Student');  
Query OK, 50 rows affected (0.031 sec)  
Records: 50 Duplicates: 0 Warnings: 0
```

```
MariaDB [fact_library]> INSERT INTO BOOK VALUES (5235, 'Beginner's Guide to JAVA', 2012, 59.95,  
'Programming', NULL), (5236, 'Database in the Cloud', 2012, 79.95, 'Cloud', NULL), (5237, 'Mastering the  
database environment', 2013, 89.95, 'Database', NULL), (5238, 'Conceptual Programming', 2013, 59.95,  
'Programming', 1229), (5239, 'J++ in Mobile Apps', 2013, 49.95, 'Programming', NULL), (5240, 'iOS  
Programming', 2013, 79.95, 'Programming', 1212), (5241, 'JAVA First Steps', 2013, 49.95, 'Programming',  
NULL), (5242, 'C# in Middleware Deployment', 2013, 59.95, 'Middleware', 1228), (5243, 'DATABASES in  
Theory', 2013, 129.95, 'Database', NULL), (5244, 'Cloud-based Mobile Applications', 2013, 69.95, 'Cloud',  
NULL), (5245, 'The Golden Road to Platform independence', 2014, 119.95, 'Middleware', NULL), (5246,  
'Capture the Cloud', 2014, 69.95, 'Cloud', 1172), (5247, 'Shining Through the Cloud: Sun Programming',  
2014, 109.95, 'Programming', NULL), (5248, 'What You Always Wanted to Know About Database, But Were  
Afraid to Ask', 2014, 49.95, 'Database', NULL), (5249, 'Starlight Applications', 2014, 69.95, 'Cloud',  
1207), (5250, 'Reengineering the Middle Tier', 2014, 89.95, 'Middleware', NULL), (5251, 'Thoughts on  
Revitalizing Ruby', 2014, 59.95, 'Programming', NULL), (5252, 'Beyond the Database Veil', 2014, 69.95,
```

```
'Database', 1229), (5253, 'Virtual Programming for Virtual Environments', 2014, 79.95, 'Programming',
NULL), (5254, 'Coding Style for Maintenance', 2015, 49.95, 'Programming', NULL));
Query OK, 20 rows affected (0.037 sec)
Records: 20 Duplicates: 0 Warnings: 0
```

```
MariaDB [fact_library]> INSERT INTO CHECKOUT VALUES (91001, 5235, 1165, '2015-03-31', '2015-04-14',
'2015-04-09'), (91002, 5238, 1209, '2015-03-31', '2015-04-07', '2015-04-05'), (91003, 5240, 1160,
'2015-03-31', '2015-04-14', '2015-04-09'), (91004, 5237, 1160, '2015-03-31', '2015-04-14', '2015-04-03'),
(91005, 5236, 1202, '2015-03-31', '2015-04-07', '2015-04-08'), (91006, 5235, 1203, '2015-04-05',
'2015-04-12', '2015-04-13'), (91007, 5244, 1174, '2015-04-05', '2015-04-12', '2015-04-08'), (91008, 5248,
1181, '2015-04-05', '2015-04-12', '2015-04-06'), (91009, 5242, 1170, '2015-04-05', '2015-04-19',
'2015-04-09'), (91010, 5235, 1161, '2015-04-05', '2015-04-19', '2015-04-05'), (91011, 5236, 1181,
'2015-04-05', '2015-04-12', '2015-04-11'), (91012, 5249, 1181, '2015-04-08', '2015-04-15', '2015-04-12'),
(91013, 5240, 1160, '2015-04-10', '2015-04-24', '2015-04-19'), (91014, 5236, 1204, '2015-04-11',
'2015-04-18', '2015-04-19'), (91015, 5237, 1204, '2015-04-11', '2015-04-18', '2015-04-13'), (91016, 5236,
1183, '2015-04-13', '2015-04-27', '2015-04-14'), (91017, 5240, 1184, '2015-04-14', '2015-04-21',
'2015-04-22'), (91018, 5236, 1170, '2015-04-14', '2015-04-28', '2015-04-14'), (91019, 5246, 1171,
'2015-04-14', '2015-04-21', '2015-04-17'), (91020, 5254, 1185, '2015-04-16', '2015-04-23', '2015-04-23'),
(91021, 5238, 1185, '2015-04-16', '2015-04-23', '2015-04-21'), (91022, 5252, 1171, '2015-04-16',
'2015-04-23', '2015-04-19'), (91023, 5249, 1207, '2015-04-16', '2015-04-23', '2015-04-19'), (91024, 5235,
1209, '2015-04-21', '2015-04-28', '2015-04-29'), (91025, 5246, 1172, '2015-04-21', '2015-04-28',
'2015-04-27'), (91026, 5254, 1161, '2015-04-21', '2015-05-04', '2015-04-26'), (91027, 5243, 1161,
'2015-04-21', '2015-05-04', '2015-04-26'), (91028, 5236, 1183, '2015-04-22', '2015-05-05', '2015-04-30'),
(91029, 5244, 1203, '2015-04-22', '2015-04-29', '2015-04-26'), (91030, 5242, 1207, '2015-04-22',
'2015-04-29', '2015-04-30'), (91031, 5252, 1165, '2015-04-23', '2015-05-06', '2015-04-30'), (91032, 5238,
1172, '2015-04-23', '2015-04-30', '2015-04-26'), (91033, 5235, 1174, '2015-04-23', '2015-04-30',
'2015-04-23'), (91034, 5240, 1185, '2015-04-23', '2015-04-30', '2015-05-01'), (91035, 5248, 1165,
'2015-04-24', '2015-05-07', '2015-04-25'), (91036, 5237, 1202, '2015-04-24', '2015-04-30', '2015-04-28'),
(91037, 5235, 1210, '2015-04-28', '2015-05-04', '2015-05-01'), (91038, 5238, 1215, '2015-04-28',
'2015-05-04', '2015-04-30'), (91039, 5240, 1222, '2015-04-28', '2015-05-04', '2015-05-06'), (91040, 5237,
1228, '2015-04-28', '2015-05-04', '2015-05-05'), (91041, 5236, 1211, '2015-04-28', '2015-05-04',
'2015-04-30'), (91042, 5235, 1220, '2015-04-29', '2015-05-05', '2015-05-05'), (91043, 5244, 1226,
'2015-04-29', '2015-05-05', '2015-05-07'), (91044, 5248, 1219, '2015-04-30', '2015-05-07', '2015-05-08'),
(91045, 5242, 1210, '2015-04-30', '2015-05-07', '2015-05-04'), (91046, 5235, 1225, '2015-04-30',
'2015-05-07', '2015-05-03'), (91047, 5236, 1218, '2015-04-30', '2015-05-07', '2015-05-07'), (91048, 5249,
1229, '2015-05-04', '2015-05-11', '2015-05-06'), (91049, 5240, 1214, '2015-05-04', '2015-05-11',
'2015-05-04'), (91050, 5236, 1220, '2015-05-08', '2015-05-15', '2015-05-13'), (91051, 5237, 1222,
'2015-05-08', '2015-05-15', '2015-05-15'), (91052, 5236, 1213, '2015-05-08', '2015-05-15', '2015-05-08'),
(91053, 5240, 1212, '2015-05-09', '2015-05-16', NULL), (91054, 5236, 1221, '2015-05-09', '2015-05-16',
'2015-05-11'), (91055, 5246, 1221, '2015-05-09', '2015-05-16', '2015-05-10'), (91056, 5254, 1224,
'2015-05-10', '2015-05-17', '2015-05-15'), (91057, 5238, 1224, '2015-05-10', '2015-05-17', '2015-05-11'),
(91058, 5252, 1171, '2015-05-10', '2015-05-17', '2015-05-15'), (91059, 5249, 1207, '2015-05-10',
'2015-05-17', NULL), (91060, 5235, 1209, '2015-05-15', '2015-05-22', '2015-05-18'), (91061, 5246, 1172,
'2015-05-15', '2015-05-22', NULL), (91062, 5254, 1223, '2015-05-15', '2015-05-22', '2015-05-16'), (91063,
5243, 1223, '2015-05-15', '2015-05-22', '2015-05-20'), (91064, 5236, 1183, '2015-05-17', '2015-05-31',
'2015-05-21'), (91065, 5244, 1210, '2015-05-17', '2015-05-24', '2015-05-17'), (91066, 5242, 1228,
'2015-05-19', '2015-05-26', NULL), (91067, 5252, 1229, '2015-05-24', '2015-05-31', NULL), (91068, 5238,
1229, '2015-05-24', '2015-05-31', NULL);
Query OK, 68 rows affected (0.028 sec)
Records: 68 Duplicates: 0 Warnings: 0
```

```
MariaDB [fact_library]> INSERT INTO WRITES VALUES (5235, 273), (5236, 383), (5237, 185), (5238, 603),
(5239, 229), (5239, 460), (5239, 592), (5240, 218), (5241, 460), (5241, 559), (5242, 581), (5243, 251),
(5244, 262), (5244, 284), (5245, 394), (5246, 251), (5247, 394), (5248, 229), (5249, 262), (5250, 383),
(5250, 438), (5251, 460), (5252, 262), (5253, 185), (5254, 559);
Query OK, 25 rows affected (0.030 sec)
Records: 25 Duplicates: 0 Warnings: 0
```

-- Questions

(1)

```
MariaDB [fact_library]> SELECT BOOK_TITLE, BOOK_COST, BOOK_YEAR FROM BOOK;
```

BOOK_TITLE	BOOK_COST	BOOK_YEAR
Beginner's Guide to JAVA	59.95	2012
Database in the Cloud	79.95	2012
Mastering the database environment	89.95	2013
Conceptual Programming	59.95	2013
J++ in Mobile Apps	49.95	2013
iOS Programming	79.95	2013
JAVA First Steps	49.95	2013
C# in Middleware Deployment	59.95	2013
DATABASES in Theory	129.95	2013
Cloud-based Mobile Applications	69.95	2013
The Golden Road to Platform independence	119.95	2014
Capture the Cloud	69.95	2014
Shining Through the Cloud: Sun Programming	109.95	2014
What You Always Wanted to Know About Database, But Were Afraid to Ask	49.95	2014
Starlight Applications	69.95	2014
Reengineering the Middle Tier	89.95	2014
Thoughts on Revitalizing Ruby	59.95	2014
Beyond the Database Veil	69.95	2014

Virtual Programming for Virtual Environments	79.95	2014
Coding Style for Maintenance	49.95	2015

20 rows in set (0.002 sec)

(2)

MariaDB [fact_library]> SELECT PAT_FNAME, PAT_LNAME FROM PATRON;

PAT_FNAME	PAT_LNAME
robert	carter
Kelsey	Koch
Cedric	Baldwin
Vera	Alvarado
Alan	Martin
Cory	Barry
Peggy	Marsh
Tony	Miles
Betsy	Malone
Nadine	Blair
Allen	Horne
Jamal	Melendez
Helena	Hughes
Jimmie	Love
Sandra	Yang
Lorenzo	Torres
Shelby	Noble
Holly	Anthony
Tyler	Pope
Thomas	Duran
Claire	Gomez
Iva	Ramos
Ollie	Cantrell
Rena	Mathis
Keith	Cooley
Jerald	Gaines
Iva	McClain
Desiree	Rivas
Marina	King
Maureen	Downs
Angel	Terrell
Desiree	Harrington
Carlton	Morton
Gloria	Pitts
Zach	Kelly
Jose	Hays
Jewel	England
Wilfred	Fuller
Jeff	Owens
Alicia	Dickson
Homer	Goodman
Gerald	Burke
Brandi	Larson
Erika	Bowen
Elton	Irwin
Jan	Joyce
Irene	West
Mario	King
Roberto	Kennedy
Leon	Richmond

50 rows in set (0.001 sec)

(3)

MariaDB [fact_library]> SELECT CHECK_NUM, CHECK_OUT_DATE, CHECK_DUE_DATE FROM CHECKOUT;

CHECK_NUM	CHECK_OUT_DATE	CHECK_DUE_DATE
91001	2015-03-31	2015-04-14
91002	2015-03-31	2015-04-07
91003	2015-03-31	2015-04-14
91004	2015-03-31	2015-04-14
91005	2015-03-31	2015-04-07
91006	2015-04-05	2015-04-12
91007	2015-04-05	2015-04-12
91008	2015-04-05	2015-04-12
91009	2015-04-05	2015-04-19
91010	2015-04-05	2015-04-19
91011	2015-04-05	2015-04-12
91012	2015-04-08	2015-04-15
91013	2015-04-10	2015-04-24
91014	2015-04-11	2015-04-18
91015	2015-04-11	2015-04-18

91016	2015-04-13	2015-04-27
91017	2015-04-14	2015-04-21
91018	2015-04-14	2015-04-28
91019	2015-04-14	2015-04-21
91020	2015-04-16	2015-04-23
91021	2015-04-16	2015-04-23
91022	2015-04-16	2015-04-23
91023	2015-04-16	2015-04-23
91024	2015-04-21	2015-04-28
91025	2015-04-21	2015-04-28
91026	2015-04-21	2015-05-04
91027	2015-04-21	2015-05-04
91028	2015-04-22	2015-05-05
91029	2015-04-22	2015-04-29
91030	2015-04-22	2015-04-29
91031	2015-04-23	2015-05-06
91032	2015-04-23	2015-04-30
91033	2015-04-23	2015-04-30
91034	2015-04-23	2015-04-30
91035	2015-04-24	2015-05-07
91036	2015-04-24	2015-04-30
91037	2015-04-28	2015-05-04
91038	2015-04-28	2015-05-04
91039	2015-04-28	2015-05-04
91040	2015-04-28	2015-05-04
91041	2015-04-28	2015-05-04
91042	2015-04-29	2015-05-05
91043	2015-04-29	2015-05-05
91044	2015-04-30	2015-05-07
91045	2015-04-30	2015-05-07
91046	2015-04-30	2015-05-07
91047	2015-04-30	2015-05-07
91048	2015-05-04	2015-05-11
91049	2015-05-04	2015-05-11
91050	2015-05-08	2015-05-15
91051	2015-05-08	2015-05-15
91052	2015-05-08	2015-05-15
91053	2015-05-09	2015-05-16
91054	2015-05-09	2015-05-16
91055	2015-05-09	2015-05-16
91056	2015-05-10	2015-05-17
91057	2015-05-10	2015-05-17
91058	2015-05-10	2015-05-17
91059	2015-05-10	2015-05-17
91060	2015-05-15	2015-05-22
91061	2015-05-15	2015-05-22
91062	2015-05-15	2015-05-22
91063	2015-05-15	2015-05-22
91064	2015-05-17	2015-05-31
91065	2015-05-17	2015-05-24
91066	2015-05-19	2015-05-26
91067	2015-05-24	2015-05-31
91068	2015-05-24	2015-05-31

68 rows in set (0.002 sec)

(4)

MariaDB [fact_library]> SELECT BOOK_NUM, BOOK_TITLE AS 'TITLE', BOOK_YEAR AS 'Year Published' FROM BOOK;

BOOK_NUM	TITLE	Year Published
5235	Beginner's Guide to JAVA	2012
5236	Database in the Cloud	2012
5237	Mastering the database environment	2013
5238	Conceptual Programming	2013
5239	J++ in Mobile Apps	2013
5240	iOS Programming	2013
5241	JAVA First Steps	2013
5242	C# in Middleware Deployment	2013
5243	DATABASES in Theory	2013
5244	Cloud-based Mobile Applications	2013
5245	The Golden Road to Platform independence	2014
5246	Capture the Cloud	2014
5247	Shining Through the Cloud: Sun Programming	2014
5248	What You Always Wanted to Know About Database, But Were Afraid to Ask	2014
5249	Starlight Applications	2014
5250	Reengineering the Middle Tier	2014
5251	Thoughts on Revitalizing Ruby	2014
5252	Beyond the Database Veil	2014
5253	Virtual Programming for Virtual Environments	2014
5254	Coding Style for Maintenance	2015

20 rows in set (0.027 sec)

(5)
MariaDB [fact_library]> SELECT DISTINCT BOOK_YEAR FROM BOOK;

BOOK_YEAR
2012
2013
2014
2015

4 rows in set (0.026 sec)

(6)
MariaDB [fact_library]> SELECT DISTINCT BOOK_SUBJECT FROM BOOK ORDER BY BOOK_SUBJECT;

BOOK_SUBJECT
Cloud
Database
Middleware
Programming

4 rows in set (0.027 sec)

(7)
MariaDB [fact_library]> SELECT BOOK_NUM, BOOK_TITLE, BOOK_COST AS 'Replacement Cost' FROM BOOK;

BOOK_NUM	BOOK_TITLE	Replacement Cost
5235	Beginner's Guide to JAVA	59.95
5236	Database in the Cloud	79.95
5237	Mastering the database environment	89.95
5238	Conceptual Programming	59.95
5239	J++ in Mobile Apps	49.95
5240	iOS Programming	79.95
5241	JAVA First Steps	49.95
5242	C# in Middleware Deployment	59.95
5243	DATABASES in Theory	129.95
5244	Cloud-based Mobile Applications	69.95
5245	The Golden Road to Platform independence	119.95
5246	Capture the Cloud	69.95
5247	Shining Through the Cloud: Sun Programming	109.95
5248	What You Always Wanted to Know About Database, But Were Afraid to Ask	49.95
5249	Starlight Applications	69.95
5250	Reengineering the Middle Tier	89.95
5251	Thoughts on Revitalizing Ruby	59.95
5252	Beyond the Database Veil	69.95
5253	Virtual Programming for Virtual Environments	79.95
5254	Coding Style for Maintenance	49.95

20 rows in set (0.001 sec)

(8)
MariaDB [fact_library]> SELECT CHECK_NUM, BOOK_NUM, PAT_ID, CHECK_OUT_DATE, CHECK_DUE_DATE FROM CHECKOUT ORDER BY CHECK_OUT_DATE DESC;

CHECK_NUM	BOOK_NUM	PAT_ID	CHECK_OUT_DATE	CHECK_DUE_DATE
91067	5252	1229	2015-05-24	2015-05-31
91068	5238	1229	2015-05-24	2015-05-31
91066	5242	1228	2015-05-19	2015-05-26
91065	5244	1210	2015-05-17	2015-05-24
91064	5236	1183	2015-05-17	2015-05-31
91060	5235	1209	2015-05-15	2015-05-22
91061	5246	1172	2015-05-15	2015-05-22
91062	5254	1223	2015-05-15	2015-05-22
91063	5243	1223	2015-05-15	2015-05-22
91056	5254	1224	2015-05-10	2015-05-17
91057	5238	1224	2015-05-10	2015-05-17
91058	5252	1171	2015-05-10	2015-05-17
91059	5249	1207	2015-05-10	2015-05-17
91055	5246	1221	2015-05-09	2015-05-16
91054	5236	1221	2015-05-09	2015-05-16
91053	5240	1212	2015-05-09	2015-05-16
91052	5236	1213	2015-05-08	2015-05-15
91050	5236	1220	2015-05-08	2015-05-15
91051	5237	1222	2015-05-08	2015-05-15
91049	5240	1214	2015-05-04	2015-05-11
91048	5249	1229	2015-05-04	2015-05-11
91047	5236	1218	2015-04-30	2015-05-07
91046	5235	1225	2015-04-30	2015-05-07
91045	5242	1210	2015-04-30	2015-05-07

91044	5248	1219	2015-04-30	2015-05-07
91043	5244	1226	2015-04-29	2015-05-05
91042	5235	1220	2015-04-29	2015-05-05
91041	5236	1211	2015-04-28	2015-05-04
91040	5237	1228	2015-04-28	2015-05-04
91039	5240	1222	2015-04-28	2015-05-04
91038	5238	1215	2015-04-28	2015-05-04
91037	5235	1210	2015-04-28	2015-05-04
91036	5237	1202	2015-04-24	2015-04-30
91035	5248	1165	2015-04-24	2015-05-07
91034	5240	1185	2015-04-23	2015-04-30
91033	5235	1174	2015-04-23	2015-04-30
91032	5238	1172	2015-04-23	2015-04-30
91031	5252	1165	2015-04-23	2015-05-06
91030	5242	1207	2015-04-22	2015-04-29
91029	5244	1203	2015-04-22	2015-04-29
91028	5236	1183	2015-04-22	2015-05-05
91024	5235	1209	2015-04-21	2015-04-28
91025	5246	1172	2015-04-21	2015-04-28
91027	5243	1161	2015-04-21	2015-05-04
91026	5254	1161	2015-04-21	2015-05-04
91023	5249	1207	2015-04-16	2015-04-23
91022	5252	1171	2015-04-16	2015-04-23
91021	5238	1185	2015-04-16	2015-04-23
91020	5254	1185	2015-04-16	2015-04-23
91017	5240	1184	2015-04-14	2015-04-21
91018	5236	1170	2015-04-14	2015-04-28
91019	5246	1171	2015-04-14	2015-04-21
91016	5236	1183	2015-04-13	2015-04-27
91015	5237	1204	2015-04-11	2015-04-18
91014	5236	1204	2015-04-11	2015-04-18
91013	5240	1160	2015-04-10	2015-04-24
91012	5249	1181	2015-04-08	2015-04-15
91006	5235	1203	2015-04-05	2015-04-12
91007	5244	1174	2015-04-05	2015-04-12
91008	5248	1181	2015-04-05	2015-04-12
91009	5242	1170	2015-04-05	2015-04-19
91010	5235	1161	2015-04-05	2015-04-19
91011	5236	1181	2015-04-05	2015-04-12
91005	5236	1202	2015-03-31	2015-04-07
91004	5237	1160	2015-03-31	2015-04-14
91003	5240	1160	2015-03-31	2015-04-14
91002	5238	1209	2015-03-31	2015-04-07
91001	5235	1165	2015-03-31	2015-04-14

68 rows in set (0.001 sec)

(9)
MariaDB [fact_library]> SELECT BOOK_TITLE, BOOK_YEAR, BOOK_SUBJECT FROM BOOK ORDER BY BOOK_SUBJECT ASC, BOOK_YEAR DESC, BOOK_TITLE ASC;

BOOK_TITLE	BOOK_YEAR	BOOK_SUBJECT
Capture the Cloud	2014	Cloud
Starlight Applications	2014	Cloud
Cloud-based Mobile Applications	2013	Cloud
Database in the Cloud	2012	Cloud
Beyond the Database Veil	2014	Database
What You Always Wanted to Know About Database, But Were Afraid to Ask	2014	Database
DATABASES in Theory	2013	Database
Mastering the database environment	2013	Database
Reengineering the Middle Tier	2014	Middleware
The Golden Road to Platform independence	2014	Middleware
C# in Middleware Deployment	2013	Middleware
Coding Style for Maintenance	2015	Programming
Shining Through the Cloud: Sun Programming	2014	Programming
Thoughts on Revitalizing Ruby	2014	Programming
Virtual Programming for Virtual Environments	2014	Programming
Conceptual Programming	2013	Programming
iOS Programming	2013	Programming
J++ in Mobile Apps	2013	Programming
JAVA First Steps	2013	Programming
Beginner's Guide to JAVA	2012	Programming

20 rows in set (0.002 sec)

(10)
MariaDB [fact_library]> SELECT BOOK_NUM, BOOK_TITLE, BOOK_YEAR FROM BOOK WHERE BOOK_YEAR='2012';

BOOK_NUM	BOOK_TITLE	BOOK_YEAR
5235	Beginner's Guide to JAVA	2012
5236	Database in the Cloud	2012


```
+-----+
2 rows in set (0.029 sec)
```

```
(11)
MariaDB [fact_library]> SELECT BOOK_NUM, BOOK_TITLE, BOOK_YEAR FROM BOOK WHERE BOOK_SUBJECT='Database';
```

BOOK_NUM	BOOK_TITLE	BOOK_YEAR
5237	Mastering the database environment	2013
5243	DATABASES in Theory	2013
5248	What You Always Wanted to Know About Database, But Were Afraid to Ask	2014
5252	Beyond the Database Veil	2014

```
+-----+
4 rows in set (0.002 sec)
```

```
(12)
MariaDB [fact_library]> SELECT CHECK_NUM, BOOK_NUM, CHECK_OUT_DATE FROM CHECKOUT WHERE
CHECK_OUT_DATE<'2015-05-05';
```

CHECK_NUM	BOOK_NUM	CHECK_OUT_DATE
91001	5235	2015-03-31
91002	5238	2015-03-31
91003	5240	2015-03-31
91004	5237	2015-03-31
91005	5236	2015-03-31
91006	5235	2015-04-05
91007	5244	2015-04-05
91008	5248	2015-04-05
91009	5242	2015-04-05
91010	5235	2015-04-05
91011	5236	2015-04-05
91012	5249	2015-04-08
91013	5240	2015-04-10
91014	5236	2015-04-11
91015	5237	2015-04-11
91016	5236	2015-04-13
91017	5240	2015-04-14
91018	5236	2015-04-14
91019	5246	2015-04-14
91020	5254	2015-04-16
91021	5238	2015-04-16
91022	5252	2015-04-16
91023	5249	2015-04-16
91024	5235	2015-04-21
91025	5246	2015-04-21
91026	5254	2015-04-21
91027	5243	2015-04-21
91028	5236	2015-04-22
91029	5244	2015-04-22
91030	5242	2015-04-22
91031	5252	2015-04-23
91032	5238	2015-04-23
91033	5235	2015-04-23
91034	5240	2015-04-23
91035	5248	2015-04-24
91036	5237	2015-04-24
91037	5235	2015-04-28
91038	5238	2015-04-28
91039	5240	2015-04-28
91040	5237	2015-04-28
91041	5236	2015-04-28
91042	5235	2015-04-29
91043	5244	2015-04-29
91044	5248	2015-04-30
91045	5242	2015-04-30
91046	5235	2015-04-30
91047	5236	2015-04-30
91048	5249	2015-05-04
91049	5240	2015-05-04

```
+-----+
49 rows in set (0.027 sec)
```

```
(13)
MariaDB [fact_library]> SELECT BOOK_NUM, BOOK_TITLE, BOOK_YEAR FROM BOOK WHERE BOOK_YEAR > 2013 AND
BOOK_SUBJECT='Programming';
```

BOOK_NUM	BOOK_TITLE	BOOK_YEAR
5247	Shining Through the Cloud: Sun Programming	2014
5251	Thoughts on Revitalizing Ruby	2014
5253	Virtual Programming for Virtual Environments	2014
5254	Coding Style for Maintenance	2015

```
+-----+-----+-----+
4 rows in set (0.027 sec)
```

```
(14)
MariaDB [fact_library]> SELECT BOOK_NUM, BOOK_TITLE, BOOK_YEAR, BOOK_SUBJECT, BOOK_COST FROM BOOK WHERE
(BOOK_SUBJECT = 'Middleware' OR BOOK_SUBJECT = 'Cloud') AND BOOK_COST > 70;
```

BOOK_NUM	BOOK_TITLE	BOOK_YEAR	BOOK_SUBJECT	BOOK_COST
5236	Database in the Cloud	2012	Cloud	79.95
5245	The Golden Road to Platform independence	2014	Middleware	119.95
5250	Reengineering the Middle Tier	2014	Middleware	89.95

```
3 rows in set (0.002 sec)
```

```
(15)
MariaDB [fact_library]> SELECT AU_ID, AU_FNAME, AU_LNAME, AU_BIRTHYEAR FROM AUTHOR WHERE AU_BIRTHYEAR
BETWEEN 1980 AND 1989;
```

AU_ID	AU_FNAME	AU_LNAME	AU_BIRTHYEAR
218	Rachel	Beatney	1983
383	Neal	Walsh	1980
394	Robert	Lake	1982
438	Perry	Pearson	1986
460	Connie	Paulsen	1983
581	Manish	Aggerwal	1984
603	Julia	Palca	1988

```
7 rows in set (0.027 sec)
```

```
(16)
MariaDB [fact_library]> SELECT BOOK_NUM, BOOK_TITLE, BOOK_YEAR FROM BOOK WHERE BOOK_TITLE LIKE
'%Database%';
```

BOOK_NUM	BOOK_TITLE	BOOK_YEAR
5236	Database in the Cloud	2012
5237	Mastering the database environment	2013
5243	DATABASES in Theory	2013
5248	What You Always Wanted to Know About Database, But Were Afraid to Ask	2014
5252	Beyond the Database Veil	2014

```
5 rows in set (0.027 sec)
```

```
(17)
MariaDB [fact_library]> SELECT PAT_ID, PAT_FNAME, PAT_LNAME FROM PATRON WHERE Pat_Type = 'Student';
```

PAT_ID	PAT_FNAME	PAT_LNAME
1166	Vera	Alvarado
1171	Peggy	Marsh
1172	Tony	Miles
1174	Betsy	Malone
1180	Nadine	Blair
1181	Allen	Horne
1182	Jamal	Melendez
1184	Jimmie	Love
1185	Sandra	Yang
1200	Lorenzo	Torres
1201	Shelby	Noble
1202	Holly	Anthony
1203	Tyler	Pope
1204	Thomas	Duran
1205	Claire	Gomez
1207	Iva	Ramos
1208	Ollie	Cantrell
1209	Rena	Mathis
1210	Keith	Cooley
1211	Jerald	Gaines
1212	Iva	McClain
1213	Desiree	Rivas
1214	Marina	King
1215	Maureen	Downs
1218	Angel	Terrell
1219	Desiree	Harrington
1220	Carlton	Morton
1221	Gloria	Pitts
1222	Zach	Kelly
1223	Jose	Hays
1224	Jewel	England
1225	Wilfred	Fuller
1226	Jeff	Owens

1227	Alicia	Dickson
1228	Homer	Goodman
1229	Gerald	Burke
1237	Brandi	Larson
1238	Erika	Bowen
1239	Elton	Irwin
1240	Jan	Joyce
1241	Irene	West
1242	Mario	King
1243	Roberto	Kennedy
1244	Leon	Richmond

44 rows in set (0.001 sec)

(18)
MariaDB [fact_library]> SELECT PAT_ID, PAT_FNAME, PAT_LNAME, PAT_TYPE FROM PATRON WHERE PAT_LNAME LIKE 'C%';

PAT_ID	PAT_FNAME	PAT_LNAME	PAT_TYPE
1160	robert	carter	Faculty
1208	Ollie	Cantrell	Student
1210	Keith	Cooley	STudent

3 rows in set (0.002 sec)

(19)
MariaDB [fact_library]> SELECT AU_ID, AU_FNAME, AU_LNAME FROM AUTHOR WHERE Au_BirthYear IS NULL;

AU_ID	AU_FNAME	AU_LNAME
229	Carmine	Salvadore
262	Xia	Chiang
559	Rachel	McGill

3 rows in set (0.001 sec)

(20)
MariaDB [fact_library]> SELECT AU_ID, AU_FNAME, AU_LNAME FROM AUTHOR WHERE Au_BirthYear IS NOT NULL;

AU_ID	AU_FNAME	AU_LNAME
185	Benson	Reeves
218	Rachel	Beatney
251	Hugo	Bruer
273	Reba	Durante
284	Trina	Tankersly
383	Neal	Walsh
394	Robert	Lake
438	Perry	Pearson
460	Connie	Paulsen
581	Manish	Aggerwal
592	Lawrence	Sheel
603	Julia	Palca

12 rows in set (0.001 sec)

(21)
MariaDB [fact_library]> SELECT CHECK_NUM, BOOK_NUM, PAT_ID, CHECK_OUT_DATE, CHECK_DUE_DATE FROM CHECKOUT WHERE Check_In_Date IS NULL ORDER BY Book_Num ASC;

CHECK_NUM	BOOK_NUM	PAT_ID	CHECK_OUT_DATE	CHECK_DUE_DATE
91068	5238	1229	2015-05-24	2015-05-31
91053	5240	1212	2015-05-09	2015-05-16
91066	5242	1228	2015-05-19	2015-05-26
91061	5246	1172	2015-05-15	2015-05-22
91059	5249	1207	2015-05-10	2015-05-17
91067	5252	1229	2015-05-24	2015-05-31

6 rows in set (0.002 sec)

(22)
MariaDB [fact_library]> SELECT AU_ID, AU_FNAME, AU_LNAME, AU_BIRTHYEAR FROM AUTHOR ORDER BY Au_BirthYear DESC, Au_LName ASC;

AU_ID	AU_FNAME	AU_LNAME	AU_BIRTHYEAR
185	Benson	Reeves	1990
603	Julia	Palca	1988
438	Perry	Pearson	1986
581	Manish	Aggerwal	1984
218	Rachel	Beatney	1983

460	Connie	Paulsen	1983
394	Robert	Lake	1982
383	Neal	Walsh	1980
592	Lawrence	Sheel	1976
251	Hugo	Bruer	1972
273	Reba	Durante	1969
284	Trina	Tankersly	1961
262	Xia	Chiang	NULL
559	Rachel	McGill	NULL
229	Carmine	Salvadore	NULL

15 rows in set (0.002 sec)

(23)

MariaDB [fact_library]> SELECT COUNT(*) AS 'Number of Books' FROM BOOK;

Number of Books
20

1 row in set (0.027 sec)

(24)

MariaDB [fact_library]> SELECT COUNT(DISTINCT Book_Subject) AS 'Number of Subjects' FROM BOOK;

Number of Subjects
4

1 row in set (0.027 sec)

(25)

MariaDB [fact_library]> SELECT COUNT(*) AS 'Available Books' FROM BOOK WHERE Book_Num NOT IN (SELECT Book_Num FROM CHECKOUT WHERE Check_In_Date IS NULL);

Available Books
14

1 row in set (0.034 sec)

(26)

MariaDB [fact_library]> SELECT MAX(Book_Cost) AS 'Most Expensive' FROM BOOK;

Most Expensive
129.95

1 row in set (0.010 sec)

(27)

MariaDB [fact_library]> SELECT MIN(Book_Cost) AS 'Least Expensive' FROM BOOK;

Least Expensive
49.95

1 row in set (0.001 sec)

(28)

MariaDB [fact_library]> SELECT COUNT(DISTINCT Pat_ID) AS 'DIFFERENT PATRONS' FROM CHECKOUT;

DIFFERENT PATRONS
33

1 row in set (0.027 sec)

(29)

MariaDB [fact_library]> SELECT BOOK_SUBJECT, COUNT(*) AS 'Books In Subject' FROM BOOK GROUP BY Book_Subject ORDER BY 'Books In Subject' DESC, Book_Subject ASC;

BOOK_SUBJECT	Books In Subject
Programming	9
Cloud	4
Database	4
Middleware	3

4 rows in set (0.003 sec)

(30)

MariaDB [fact_library]> SELECT Au_ID, COUNT(Book_Num) AS 'Books Written' FROM WRITES GROUP BY Au_ID ORDER

BY 'Books Written' DESC, Au_ID ASC;

+-----+-----+	
AU_ID	Books Written
+-----+-----+	
262	3
460	3
185	2
229	2
251	2
383	2
394	2
559	2
218	1
273	1
284	1
438	1
581	1
592	1
603	1
+-----+-----+	

15 rows in set (0.003 sec)

(31)

MariaDB [fact_library]> SELECT SUM(Book_Cost) AS 'Library Value' FROM BOOK;

+-----+	
Library Value	
+-----+	
1499.00	
+-----+	

1 row in set (0.002 sec)